# Airflow Is the Load in Heating and the Delivery in Cooling

One Asymmetry Decides Three Controller Decisions in the Smart Buildings Challenge

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**Abstract**

The Smart Buildings Challenge asks whether a single HVAC control policy can transfer to buildings and climates it has never seen, and poses as an open question whether domain-specific models or generic foundation models will generalize best. We report a result on the domain-specific side, and a caution about how it should be measured.

One property of the plant decides three separate controller choices, and the shipped reference controllers get all three wrong in the same direction. The air handler cannot heat—all heat comes from the VAV reheat coils—and below about $-6.8^{\circ}$C ambient its supply-air command is inert as well, because it delivers $\min(\text{mixed air}, \text{setpoint})$ and with recirculation capped at 70% the mixed air is already below the coldest setpoint available. So **in heating, airflow is the load**: three tenths of every cubic metre the fan moves is ambient air that must then be heated, and the fan work is spent before the gas is spent undoing it. **In cooling the same airflow is the delivery.** Every decision that adds or withholds airflow therefore pays in one direction and costs in the other, and each needs the same gate: static pressure, air-handler run length, and whether to run a handler at all.

Acting on this takes the metric from 0.05496 to 0.02774 over the 30 published scenarios, a 49.5% reduction, with no learned parameters and in pure NumPy. An earlier version of the controller measured 0.03785 locally and scored 0.04932 on the organizers' held-out leaderboard against 0.06404 for the baseline, so the gain transfers at 23% and the two scales are readable against each other.

We also report eight refuted hypotheses, including our own initial one and one conclusion we drew and then had to withdraw: after six failed attempts at the comfort term we claimed the shipped gains sat near a local optimum. They do not. Every one of those attempts pushed the zone loop *harder*, and the loop was already oscillating; the fix is a five-times softer gain with an integral term, which no single-knob ablation can find because neither half works alone.

## 1 Introduction

The challenge [1] evaluates one control policy on six unseen buildings at five unseen weather stations. Buildings differ in floor plan, zone count, air-handler count and envelope construction, so both the observation and action vectors change shape per building and there is no fixed layout to memorize. The organizers state the core hurdle as transferability, and ask whether "domain specific models will win out, or whether generic foundation models will generalize best" [1].

Two features of the setup push against learned policies. Per-rollout wall-clock is bounded at ten times the `bang_bang` baseline, which rules out winning by online planning. And the metric is a mean over 30 equally weighted scenarios whose difficulty is very unequal: in our measurements the four worst scenarios contribute 38% of the strongest baseline's total loss. Mean-case improvement is worth a fraction of tail-case improvement, and tails are where sample-efficient learning is weakest.

This paper takes the other route. We read the reward function and the plant model as a price list, find one actuator whose price the reference controllers have the wrong sign for, and correct it.

## 2 The reward as a price list

Per five-minute step the reward is

$$r_t = 0.6\,p_t - 0.2\,c_t - 0.2\,e_t, \tag{1}$$

with all three terms penalties, $r_t \le 0$, and the leaderboard metric the mean over scenarios of $|\bar{r}|$. Substituting the simulator's own cost and carbon models at the shipped normalisation constants (320 kW electricity, 820 kW gas) gives the marginal cost in metric units of one delivered kilowatt-hour:

| | metric cost per kWh |
|---|---|
| fan / pump electricity | 0.0064 |
| gas heat (boiler, $\eta \approx 1$) | 0.0034 |
| chiller cooling (COP 3.5) | 0.0018 per kWh *thermal* |

Electricity costs 1.9× gas per kWh of fuel, and a fan has no coefficient of performance to divide by where a chiller has 3.5. **Moving air is the most expensive thing this plant can do.**

The comfort term is an occupancy-weighted mean of per-zone shortfall, where shortfall is dev/5 K inside the knee and quadratic and unbounded beyond it. Every occupied zone sitting 1 K outside its band therefore costs $0.6 \times 0.2 = 0.12$ per step, which at 0.0064 per kWh is 19 kWh per step—225 kW held continuously, against a 320 kW plant. *While comfort is scored, energy is nearly free beside it.*

Finally, the comfort term is gated hard on occupancy: `total_occupancy > 0.0` or the term is identically zero whatever the zone temperatures are. Occupants arrive between hours 6 and 12, leave between 13 and 19, and are absent on weekends and holidays, leaving roughly 60% of the 8,640 scored steps with the comfort term switched off.

## 3 Airflow is the load

The gated comfort term invites an obvious move: stop conditioning an empty building. We implemented it and it was *worse* than the strongest baseline on every scenario we tried. Section 6 gives the two reasons. The useful finding came from asking instead why the reference controllers spend what they spend.

Every policy in the shipped `pid` family sets static pressure from the control effort $u$:

$$\texttt{pressure} = \mathrm{clip}(2|u| - 1,\ -1,\ 1). \tag{2}$$

The baseline's own documentation defends this against pinning pressure at maximum, and against that alternative it is an improvement. But $|u|$ does not distinguish the two directions of the loop, and the directions are not symmetric.

**Heating.** The air handler cannot heat. It mixes return and outside air, cools the mixture towards the commanded supply temperature, and all heat in the building comes from the VAV reheat coils fed by the hot-water loop. In cold weather the supply-air action is inert as well: the handler delivers min(mixed air, setpoint), and with recirculation capped at 0.70, return air near 293 K and ambient at 231 K,

$$\text{mixed} = 0.70 \times 293 + 0.30 \times 231 = 274 \text{ K}, \tag{3}$$

already below the 285 K floor the setpoint can command.

So in heating the fan moves 274 K air past a coil, and every cubic metre of it must be heated from 274 K before it does the zone any good. The ventilation floor makes this unavoidable: three tenths of the moved air is ambient, at −42°C in the coldest shipped strip. **The fan work is spent, and then gas is spent undoing the air the fan moved.**

The thermal energy a zone needs is fixed by its heat loss. It can arrive as a lot of slightly warm air or a little hot air, and the second is strictly cheaper here, because the gas is the same either way and the fan is not. Equation (2) does the opposite: it buys more of the expensive fuel in order to import more of the load.

**Cooling.** Air is the only delivery mechanism, so flow must be bought. But the trade still favours temperature over flow—the chiller kilowatt-hour is a third of the fan kilowatt-hour—and the same ventilation floor means that in 45°C ambient, extra flow again imports extra load.

# 4 Controller

Our policy is the shipped `pid_water` controller with four changes and no learned parameters. Three of the four are the same asymmetry applied to a different actuator.

**A flat band.** `pid_water` widens its band to an eco setback outside comfort hours. We hold the comfort band around the clock. This is not an energy argument but a tail argument: a building held in band overnight starts the morning with no deviation to recover, and the recovery penalty is larger than the holding cost. The shipped `fixed_setpoint` already hints at this by beating `pid_water` in cold climates.

**A direction-aware pressure law.** Where (2) applies, we instead clamp static pressure to its floor whenever the handler is heating ($u > 0$), and scale it by 0.6 when cooling. The simulator enforces a minimum static pressure under any running handler, so the floor requests that minimum rather than no air; ventilation is not something a policy may trade away, and this does not try to.

One further change follows from the same reading. `pid_water` opens the reheat valve in proportion to shortfall, fully open at 2 K. That protects a mild climate from overshoot and starves a cold one, where the valve is the only heat source and the air reaching it is already 20 K below the zone. We saturate the valve at 0.5 K.

**A minimum run length for cooling runs.** The shipped loop idles a handler the instant its *aggregate* zone error clears and restarts it at the next excursion. Because the zone loop oscillates (below), those excursions arrive every few steps: every handler switches on or off 50 to 107 times a day. Each restart is paid for twice—the fan spins up against a cold duct, and the zones drift while it is off, so the restart is larger than the excursion that triggered it. We hold a run that was started to cool at its last commands for 288 steps.

Heating runs are not held, and that is the asymmetry again: a handler that cannot heat has no reason to stay on, and holding one in the cold is a continuous flow of 274 K air for nothing. We also tested forcing every handler on through the comfort window—what a real VAV plant does, supplying cool air steadily while each box decides how much to take. Gated on the zones' own demand it is worth −44% on `phoenix_4` and exactly 0.0% on `fairbanks_1`, because the gate never fires there. Ungated it costs +54% in Fairbanks. But combined with the hold it is worse than the hold alone (0.0506 against 0.0385): both mechanisms exist to stop a handler being cycled off, so together they interfere. We kept the hold.

**Per-zone PI against the band centre.** The shipped zone law is

$$\texttt{damper} = \min\left(1, \frac{|\texttt{deadband_error}|}{2\,\mathrm{K}}\right), \tag{4}$$

proportional only, and aimed at the band *edge*: `deadband_error` is zero everywhere inside the band, so the loop has no target in there to hold. It pushes until the zone crosses the edge, stops, and the zone drifts back out the other side. Zones average *inside* the 21–24°C band and yet sit outside it 57–61% of working hours, swinging 2–4 K every few steps [3].

We give each zone its own PI loop on $e = T - \text{centre}$, with $k_p = 0.1$, $k_i = 0.005$ per five-minute step and conditional integration, from an hour before the comfort window until it closes. One signal drives both actuators: the damper follows $|u|$, because air is needed either to cool or to carry the reheat coil's output, and only the valve follows the sign.

# 5 Results

All runs are full 8,640-step episodes. The simulator is deterministic—we verified two runs of one policy on one scenario reproduce to ten decimal places—so each number is a single rollout and no averaging is required.

Table 1: Leaderboard metric (lower is better) over the 30 published training scenarios, with the weighted contribution of each reward term. Both policies re-run with the current simulator.

| policy | metric | worst scenario | comfort | energy | carbon |
|---|---|---|---|---|---|
| `pid_water` (strongest baseline) | 0.05496 | 0.21392 | 0.03489 | 0.01150 | 0.00857 |
| band + fan clamp only | 0.03785 | 0.07262 | 0.02655 | 0.00678 | 0.00452 |
| **+ zone PI + cooling hold** | **0.02774** | **0.07685** | 0.01576 | 0.00706 | 0.00492 |
| change | −49.5% | −64.1% | −54.8% | −38.6% | −42.6% |

**The fan clamp improves all three terms at once**, which is the signature of a mispriced actuator rather than a trade: less static pressure is less airflow, which is less ambient air imported, which is less load, so the coils keep up where before they could not. Comfort improves *because* the energy was cut, not despite it.

The zone PI and the hold then take comfort down by a further 41% for a slight rise in energy, which is the trade the 35:1 price ratio says to make.

**The remaining tail is now entirely one climate.** The four worst published scenarios are `fairbanks_1`, `fairbanks_7`, `fairbanks_8` and `fairbanks_2`, at 0.077, 0.067, 0.061 and 0.060. The held-out `dev` split says the same thing across independent buildings, by station:

| | atlanta | phoenix | singapore | denver | fairbanks |
|---|---|---|---|---|---|
| `dev` score | 0.0137 | 0.0137 | 0.0152 | **0.0655** | **0.0644** |

The two cold stations are four to five times the other three. That is the design showing through rather than a surprise: both of the later changes are gated on cooling, so neither fires in the cold, and what remains is exactly the climate they do not touch. It is also where the asymmetry says the problem is hardest—in heating the one actuator that moves air is working against the load it is trying to serve, so there is no cheap direction to push.

The gain is concentrated in the tail, which is what the metric rewards. Our policy is better on 19 of 30 scenarios; summed over scenarios it improves 0.575 and regresses 0.031, a ratio of 19:1, and the median regression among the eleven is 0.001. The four largest improvements are `fairbanks_1` 0.214→0.072, `phoenix_4` 0.160→0.043, `phoenix_5` 0.136→0.049 and `singapore_4` 0.110→0.024: the two coldest and the two hottest, which is what a mechanism acting through the ventilation floor predicts.

Table 2: Ablation, mean over `fairbanks_1`, `singapore_6` and `phoenix_4`. Each row adds one change to the row above.

| | metric | |
|---|---|---|
| `pid_water` | 0.13478 | shipped baseline |
| + flat band | 0.12321 | band alone |
| + cooling clamp only | 0.11315 | |
| + heating clamp only | 0.05226 | **the single biggest term** |
| + reheat saturation at 0.5 K | 0.05034 | |
| + per-zone PI | 0.04657 | |
| + cooling-run hold, 288 steps | **0.03846** | final |
| *tested and dropped* | | |
| PI, proportional only ($k_i = 0$) | 0.05304 | worse than no PI at all |
| PI + forced-on handlers | 0.04248 | |
| PI + forced-on + hold | 0.05061 | the two interfere |

### 5.1 Held-out buildings and climates

Because the published 30 scenarios are a training set, we built two replacements from material they never use. The generator is procedural, ten of the 48 shipped weather strips are unused by the published set, and a further station, `minneapolis`, is used by none of them. **dev** is four newly generated buildings crossed with five unused strips at seen stations; **test** is four further buildings crossed with `minneapolis`, so neither the building nor the climate has been seen. All eight fall inside the 28–85 zone and 3–9 air-handler range the organizers state for the evaluation set.

Table 3: The same controller on three sets. The gain is positive everywhere and much smaller off the published scenarios.

| | `pid_water` | fan clamp | full | change |
|---|---|---|---|---|
| published (30 scenarios) | 0.05496 | 0.03785 | 0.02774 | −49.5% |
| dev (20, unseen buildings) | 0.04630 | 0.04336 | 0.03449 | −25.5% |
| test (8, unseen also climate) | 0.11253 | 0.09781 | 0.08937 | −20.6% |
| **held-out leaderboard** | **0.06404** | **0.04932** | — | **−23.0%** |

The last row is the organizers' own held-out set, obtained by submitting the unmodified `pid_water` baseline alongside our entry. A baseline whose local value is known exactly turns the leaderboard into a ruler: it scores 1.17× *higher* there than on the published scenarios, so the evaluation set is slightly harder rather than easier, and the 23% gain is measured on buildings and weather we never saw.

Our `test` split is the weakest of the three estimates and we would not lean on it: it is entirely `minneapolis`, a cold station, and both of the later changes are gated on cooling. It measures the one climate they do not address, which makes −20.6% a lower bound rather than a representative figure.

**We think the gap is the finding, not a flaw in it.** The worst-case column explains it: `pid_water`'s worst published scenario scores 0.214, its worst dev scenario 0.089. The published 30 contain catastrophic baseline failures—`fairbanks_1`, `phoenix_4`, `phoenix_5`, `singapore_4`—and our change is largely repairing those. Where the baseline does not fail badly there is less to win. A method evaluated only on the published set would therefore report roughly five times the improvement it should expect on fresh buildings from the same generator.

The decomposition shifts too. On dev, energy and carbon still fall sharply ($0.0073 \to 0.0053$ and $0.0071 \to 0.0052$) but comfort rises slightly ($0.0318 \to 0.0332$), where on the published set all three improved. `phoenix` is the only station that regresses, at +2.9%, which points at the cooling clamp rather than the heating clamp as the term trading comfort away—consistent with the mechanism, since in cooling the air is the delivery path and not only the load.

We note one way our holdout may differ from the organizers'. The shipped floor generator's default footprint is 400–700 × 300–500 ft, which produces buildings far outside the stated evaluation range—one seed gave 162 zones and 17 air handlers. We therefore drew footprints from the shipped six's own range and rescaled until the realised zone count landed in 28–85. If the organizers' held-out buildings were drawn differently, our estimate of the transfer gap moves with them.

## 6 What did not work

We report these because each was a reasonable inference from the reward and each was wrong, which is itself information about the benchmark.

**Switching the plant off when the building is empty.** Our original hypothesis, and the sign was backwards. Two independent reasons. First, being empty *inside* the scored window is not an opportunity: arrivals are independent stochastic draws, so the building is routinely empty for a few steps after hour 6, and because regret is normalized by *total* occupancy the first person through the door carries their zone's entire comfort score alone. Second, a controller that coasts must later recover, and the recovery lands in the low-occupancy morning where each occupant's weight is largest. Driving to the comfort band overnight to

avoid that cost *more* than coasting saved: unoccupied-step loss rose from 447 to 830 for a comfort gain of 0.010.

**Deeper setbacks.** Monotonically worse: 0 K 0.123, 3 K 0.166, 5 K 0.160, 8 K 0.183. Steady holding beats cycling here.

**Aiming inside the comfort band.** The penalty is flat zero anywhere inside the band, so a margin looked like free insurance against drift. It is clearly harmful (0.236 against 0.123), and harmful on the *comfort* term, not merely the energy term. Narrowing a 3 K band to 2 K leaves the loop, which saturates at $1/k_p = 2$ K, oscillating across the real edge.

**Forcing the water loop hotter.** Having removed the airflow, raising the water temperature to compensate looked necessary. It is not: carbon rose from 0.0134 to 0.0162. With the flow down, the baseline's own demand reset is already hot enough.

**Suspending the fan economy when comfort is at stake.** Against the 35:1 price ratio this should pay—a quiet fan delivers heat slowly, and slow is only cheap when nobody is being charged. It does not pay (0.074 against 0.050), and the reason is the mechanism itself: the fan is what *causes* the load, so spending the expensive fuel to deliver heat imports more cold air than the extra flow delivers.

**Capping the hot-water pump.** Also electricity, also pinned at maximum by the baseline, and also worse when capped (0.051 against 0.050).

**Everything aimed at the comfort term, and a conclusion we withdrew.** After the fan clamp, comfort is 70–75% of what is left on all three sets. Against a reference of 0.0503 on the three hard scenarios we measured: reading each zone's band from its own setpoint sensors 0.0548; a stiffer loop, $k_p$ $0.5 \to 1.0$, 0.0794; dampers opening at 1 K instead of 2 K 0.0897, and at 0.5 K 0.1117; driving the air-handler loop from its worst zone rather than the mean 0.0970; a per-zone damper gate withholding airflow from a zone the supply air cannot help 0.0626; and a margin aiming inside the band rather than at its edge, 0.236.

Six attempts, all worse, and we concluded that the shipped gains sat near a local optimum and the comfort term was not reachable by retuning. **That conclusion was wrong**, and it is worth reporting why, because the error is structural rather than careless.

Four of the six—margin, $k_p$, damper gain, worst-zone—are pushes in the direction that *causes* the oscillation, applied to a loop that was already oscillating. The correct direction was five times softer. But a softer proportional gain alone is also worse: measured, $k_p = 0.1$ with $k_i = 0$ scores 0.0530 against 0.0503 for leaving the loop alone, because soft P trades the swing for a standing offset. The gain and the integral only work together, and aiming at the band centre rather than the edge is required for the integral to have a target at all.

**A one-factor-at-a-time ablation cannot find a change whose two halves are each harmful alone.** That is the methodological lesson, and six consistent negative results made it look like evidence about the problem when it was evidence about the search.

Two of the six do answer questions the starter kit leaves open: it describes both the sensor band and the worst-zone loop in its documentation and ships with both switched off. Our measurements say it was right to.

# 7 A note on the shipped baseline table

The starter kit ships `baseline_returns.json`, and `score_leaderboard.py` prints it beside a participant's score so the four baselines need not be re-run. We find it no longer reproduces: `pid_water` on `building_train_1/fairbanks_1` gives 0.21392 with current code against the recorded 0.21168, and 0.05496 against 0.05414 over the 30 scenarios.

The file was committed with the initial starter kit and never regenerated, while a later commit fixed the simulation tracker querying occupancy with VAV ids—a fix whose own message notes that it "creates fictitious populations and consumes random draws, changing subsequent simulation trajectories". The trajectories

moved and the table did not. The discrepancy is 1.1–1.5%, which is small but is the same order as several of the tuning effects above. Participants comparing against the shipped table are measuring against a simulator that no longer exists.

# 8 Discussion

On the organizers' open question, this is evidence that domain structure is worth a good deal here, and cheaply. The result is a 40-line change, dimensionless, with no fitted constants and nothing to transfer, and its sign is the same on every set we measured because it acts through a constraint—the 70% recirculation cap—that is a property of the plant rather than of any building.

The headroom here divides into two kinds, and they are found in different ways. One is *pricing*: the controller buying the most expensive fuel in order to import the load it was trying to remove. That one is visible by reading the plant and the reward, and it is what the heating-side fan clamp corrects. The other is *tracking*: a zone loop oscillating across the band it is meant to hold, for 57–61% of working hours. That one is visible only by logging what the controller actually does, step by step, which is how it was found [3] and not how we were looking.

We spent six experiments concluding the second kind did not exist, on the strength of six failures that were all the same mistake. The honest summary is that reading the reward told us where the money was, and it was not sufficient to tell us how to collect it.

Its *magnitude*, though, is a property of the evaluation set, and that is the part we would most want other entrants to take from this. The difference between 33% and 5.7% is not noise and not overfitting in the usual sense—nothing was tuned on the published scenarios that is not also present on dev—but a difference in how often the baseline's failure mode occurs. Any entry that reports only published-set numbers is reporting the frequency of those failures as much as the quality of its method.

We would not read it as evidence against learned policies in general. We would read it as a statement about what a learned policy would have to discover: not a smooth function of the observation, but the sign of a coupling between an actuator and the load it is meant to serve, visible only in the regime where the supply-air action has gone inert. A policy-gradient method optimizing an 8,640-step return at ten times `bang_bang`'s runtime budget has a hard road to that. The natural synthesis is to let structure carry the mechanism and let learning carry what structure cannot express—which is the direction we intend for the final submission.

**Limitations.** The tuning in Table 2 beyond the heating clamp is small, measured on three scenarios, and we do not claim those constants transfer. The cooling clamp in particular is a wash once measured properly: dropping it moves the score by +2.7% on the published set, −0.7% on dev and +0.3% on test, which averages to 0.6% and is not a difference we would defend. **The heating clamp is the result; the rest is noise around it.** Our policy is better on 19 of 30 published scenarios and only 8 of 20 dev scenarios; it wins on the mean because the losses are tiny (median 0.001) and the wins are large, which is the right shape for this metric but is not the same as being uniformly better. Our held-out buildings are our own construction and may not match the organizers' draw. We have not run against the organizers' held-out set, and the public leaderboard scores only a rotating subset of it.

**Reproducibility.** Code, every sweep in this paper as JSON, and the held-out generator are at `https://github.com/MinhNhat-2504/hvac-generalization-lab`. Each negative result above is a registered policy variant, runnable by name.

# References

[1] J. Goldfeder, A. Rai, P. M. Wyder, N. Li, V. Dean, D. Vakalis, J. Lin, B. Dong, M. Rossetti, N. Kutz, S. Arisaka, J. Sipple, H. Lipson, and Y. Bengio. The Smart Buildings Challenge: Learning Foundation Models for Real-World Optimization at Scale. arXiv:2610.18511, 2026.

[2] Smart Buildings 2026 Organizers. Smart Buildings Competition Starter Kit. https://github.com/smart-buildings-2026/starter-kit, 2026.

[3] Smart Buildings: per-zone PI control and AHU cooling hold. Public competition notebook, Kaggle Code tab, 2026. The per-zone oscillation diagnosis, the PI gains and the cooling-run hold are from this notebook; our implementations are independent. It also reports the minimum-fan-pressure result we arrived at separately, with the same explanation, which we take as mutual confirmation of the mechanism.